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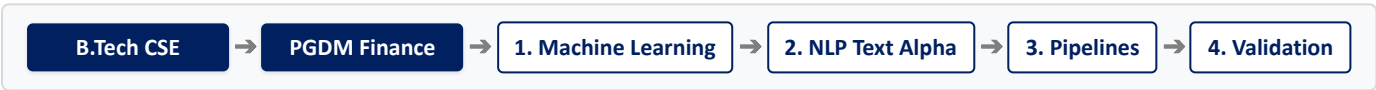
EXECUTIVE SNAPSHOT

An AI/ML Analyst combining a strong foundation in Computer Science and Finance to translate unstructured narratives and complex data points into actionable insights. I specialize in building interconnected analytical systems that treat structured asset returns and qualitative news trends as a single cohesive ecosystem. My expertise spans predictive modelling, natural language processing (NLP), reusable view-based database architectures, and empirical framework validation. Rather than implementing standalone statistical models, I build automated pipelines designed to eliminate manual spreadsheet errors and preserve operational reliability across shifting market conditions.

WHY THIS PORTFOLIO?

This portfolio presents a unified, connected, and Scrum-driven analytical framework rather than showcasing isolated academic projects. Each of the four stages was designed to solve a specific limitation observed in the previous step, establishing a clear line of technical development. This document serves as transparent proof of my ability to write production-grade SQL/Python, design reusable database pipelines, and validate model behaviour in a structured financial environment.

MY AI & DATA ANALYTICS JOURNEY



CAPABILITY MATRIX

DATA ANALYTICS	MACHINE LEARNING	NATURAL LANGUAGE PROCESSING	BUSINESS ANALYTICS
- SQL / SQL Views - Google BigQuery - Power BI - Microsoft Excel	- Machine Learning - XGBoost - Random Forest - Predictive Analytics - Model Evaluation	- spaCy / VADER - Sentiment Analysis - Named Entity Recognition - Theme Identification - Text Analytics	- Financial Analysis - Business Analysis - Data Validation - Exploratory Data Analysis

CORE ARCHITECTURAL PRINCIPLES

- Modular Architecture:** The project was designed by separating data storage, data processing, and analytical calculations into different layers. This made the workflow easier to manage, reduced errors, and allowed new data to be added without affecting the existing system.
- Reliability-Driven Testing:** The models were evaluated using multiple performance measures, including the Confusion Matrix, Accuracy, Precision, Recall, and comparisons between training and test results. High-confidence wrong predictions were also analyzed to assess model reliability, ensuring the models generalized well to new data instead of simply memorizing historical patterns.
- Agile Execution:** The project was divided into small, well-defined stages, with each stage focusing on a specific objective. Using view-based checkpoints in Google BigQuery made the workflow easier to organize, monitor, and validate throughout the project.

PROFESSIONAL PHILOSOPHY

While models and algorithms continue to evolve, structured analytical thinking, deep business context, and the development of stable, reusable systems remain the true foundations of financial data value.

PORTFOLIO PROJECTS

The following projects outline a step-by-step engineering journey where each stage validates the logical progression of the analytical framework.

Project 1: Machine Learning for Financial Market Analysis

SQL • Google BigQuery ML • Supervised Classification

Question: Can we use historical stock return data to build stable predictive models under changing market conditions?

Approach: Developed regression and classification models in BigQuery ML using returns of five major stocks to predict Nifty 50 movement. Analyzed residuals and errors (MAE, RMSE) compared to index standard deviation.

Key Finding: Shifted focus from exact return values to direction prediction. A Logistic Classification model minimized High-Confidence Wrong predictions, proving much more stable than complex non-linear models.

Project 2: Natural Language Processing for Financial Text Analytics

Python • spaCy • VADER Sentiment Analysis

Question: Can we extract structured signals from unstructured news narratives to explain market movements?

Approach: Built a Python NLP pipeline to parse financial news using VADER sentiment scoring and spaCy NER. Grouped extracted noun-chunk phrases into broader thematic categories (e.g., Risk & Uncertainty, Economic Growth).

Key Finding: Sentiment tone alone was a weak signal (50% alignment). However, thematic grouping successfully reduced noise, and combining earnings sentiment with market themes yielded a 72.37% alignment with actual stock returns.

Project 3: Reusable Analytical Systems Design

SQL • Google BigQuery Views • Ingestion Pipelines

Question: How can we convert manual and static financial models into reusable, dynamic database workflows?

Approach: Restructured the ML and NLP workflows inside BigQuery SQL using a layered, dependency-connected view architecture. Developed a reusable Python-based news extraction script using Google News RSS and feedparser.

Key Finding: Building workflows around SQL views allows the downstream analysis to update automatically when raw tables are refreshed, removing manual spreadsheet formatting risks.

Project 4: Analytical Framework Validation

BigQuery SQL • Out-of-Sample Validation Pipeline

Question: Do the reusable workflows maintain their original analytical behaviour as new market data is appended?

Approach: Validated framework stability by appending six months of unseen out-of-sample data. Evaluated model metrics (Accuracy, Precision, Recall) and compared signal distributions without retraining the models.

Key Finding: The updated validation showed that the ML classification models maintained stable performance metrics without any major stability issues, while the NLP signal analysis adapted to new market topics with some variations. Overall, the reusable framework proved highly reliable for ongoing analysis.

PORTFOLIO OUTCOMES

- Designed and validated a complete 4-stage analytical system connecting quantitative market returns with unstructured qualitative text.
- Automated data ingestion processes using dynamic, layered BigQuery views to replace fragile spreadsheet dependencies.
- Built an NLP processing framework that achieves a **72.37% alignment with actual stock returns** by combining corporate earnings sentiment and market themes.
- Verified the Logistic model achieved the **lowest out-of-sample high-confidence wrong rate of ~5.0%**, providing superior operational safety.
- Validated complete framework stability by **ingesting six months of unseen out-of-sample market observations** without requiring model retraining.

CAREER FOCUS

Seeking target opportunities as an **AI/ML Analyst or Data Analyst**. Equipped to combine technical computer science skills with corporate financial fluency to deploy reusable data products, protect database pipeline integrity, and deliver clear decision support across core business operations.

EXPLORE THE COMPLETE PORTFOLIO

GitHub Portfolio: github.com/Siddhartha-Jain-Portfolio

Please visit my GitHub profile to find my working analysis showing how I completed the projects step-by-step, along with full project reports, custom SQL schemas, and the datasets used.